

Step card: run, read, one new thing at a time

Three windows from the repo root. **A** the pretend plant system. **B** your server. **C** the checks. No typing: each step has a ready-made file. **Behind?** Copy the next step file over yours and carry on (it is a complete server): copy `services\mcp_server_reference\steps\step3_more_reads.py` `s26_server.py` (Mac or Linux: `cp`).

Window	Command	When
A	<code>uvicorn services.mock_erp.main:app</code>	Once, before step 1. Restart once in step 5.
C	<code>python -m services.mock_erp.tour</code>	Step 0. Should end: All 29 replies had the expected status.
B	<code>python s26_server.py</code>	After each step: Ctrl+C, up-arrow, Enter.

The eight steps

#	What it adds	Run in window C	You should see
1	One tool, one server	<code>python -m services.mcp_server_reference.call discover</code> <code>python -m services.mcp_server_reference.call get_equipment tag=P-1201A</code>	A server named oq-erp-mcp 0.1.0, then a pump record.
2	Inputs are checked	<code>python -m services.mcp_server_reference.call get_equipment tag=pump-one --brief</code>	A bad tag is refused before the plant system is called.
3	Three read-only tools	<code>python -m services.mcp_server_reference.call list --brief</code>	Three read tools in an ordered, cacheable list.
4	One log line per call	<code>python -m services.mcp_server_reference.call get_equipment tag=X-0000 --brief</code> <code>python -m services.mcp_server_reference.call delete_everything --brief</code>	A line in the audit log each time, including the refused one.
5	Secrets from the environment	(restart window A and B, then repeat a call)	The key is never in code or log.
6	The one write needs a person	<code>python -m services.mcp_server_reference.call raise_work_order --args-file facilitator\prebaked_outputs\mcp_server\work_order.json --brief</code>	An approval request first, then the result once approved.
7	Try to get round the gate	<code>...call raise_work_order --args-file ...\work_order_2.json --brief --approve intern:bob --reason "looks fine"</code>	Every attempt refused, and each one logged.
8	Prove it	<code>python services/mcp_server_reference/test_inspector.py --module s26_server</code>	62 checks, all green.

Two questions before the break

1. Where in the server did the bad request get stopped, and before what? 2. Which log line proves it, and what does it say?

Output looks different from the screen? Say so. Prebaked output for every step is in

`facilitator/prebaked_outputs/mcp_server/`. Commands use Windows paths; on Mac or Linux use forward slashes.